

Tejaswini Medi
Ph.D. Candidate in Machine Learning * Generative Models & Computer Vision
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Profile

Ph.D. candidate (expected completion early 2027) in the Machine Learning Group at the University of Mannheim, specialising in high-fidelity generative modelling for 2D images and 3D shapes and in adversarial robustness and fairness of deep neural networks. First author of papers at ECCV, BMVC, WACV, GCPR and an ICCV workshop (oral). Industry research experience at Autodesk Research and Endress+Hauser, both resulting in peer-reviewed publications. Seeking Research Scientist / Applied Scientist roles in generative AI and computer vision.

Research Focus

Generative Modelling	Autoregressive & transformer-based generation, wavelet / frequency-aware image and 3D shape synthesis, implicit 3D representations, flow matching
Trustworthy ML	Adversarial training, class-wise robustness, fairness under adversarial attacks, RL for training optimisation
Representation Learning	Hierarchical (coarse-to-fine) representations for 2D and 3D data

Research Experience

Doctoral Researcher 2023 - Present

Machine Learning Group, University of Mannheim

Mannheim, Germany

- * Proposed a frequency-decoupled image generation approach that separates high- and low-frequency content for faithful fine-detail synthesis (ECCV 2026).

- * Designed targeted and mixed adversarial training methods (FAIR-TAT, TRIX) and an RL-based method (RL-FAT) that improve class-wise fairness of robust models (WACV 2025, BMVC 2026).

- * Built reproducible PyTorch / PyTorch Lightning codebases with distributed training (Ray) on multi-GPU HPC clusters.

Research Intern

Jul 2024 - Sep 2024

Autodesk Research, AI Lab

- * Developed 3D-WAG, a hierarchical autoregressive 3D generative model built on wavelet-based representations that captures coarse-to-fine shape detail.

- * Achieved fast inference while producing high-fidelity 3D shapes; work published at BMVC 2025.

- * Bridged academic research with industrial 3D design workflows in collaboration with Autodesk scientists.

Student Researcher

Mar 2022 - Sep 2022

Endress+Hauser Group

Germany

- * Developed FullFormer, a transformer-based autoregressive implicit generative model that handles both watertight and non-watertight surfaces, enabling generation of nested 3D shapes.

- * Published at DAGM GCPR 2023; invited talk at SFI Visual Intelligence, Norway.

Student Researcher

Jan 2021 - Oct 2021

University of Siegen

Siegen, Germany

- * Applied policy-based reinforcement learning to facility layout planning, minimising material-handling costs.

- * Supported an RL system for dynamic flexible job-shop scheduling and built its user interface; developed a web crawler for literature references.

Education

Ph.D. in Machine Learning

2023 - Early 2027 (expected)

University of Mannheim * Advisor: Prof. Dr.-Ing. Margret Keuper

Mannheim, Germany

M.Sc. Mechatronics

2019 - 2022

University of Siegen

Siegen, Germany

B.E. Mechanical Engineering, with Distinction

2014 - 2018

Chaitanya Bharathi Institute of Technology (CBIT), Osmania University

Hyderabad, India

Publications

Full list: Google Scholar

T. Medi, H.-Y. Wang, A. Rampini, M. Keuper. Decoupling High and Low Frequencies for Faithful Image Generation with Fine Details. ECCV 2026

T. Medi, L. Mikeladze, M. Keuper. RL-FAT: Reinforcement Learning for Fair Adversarial Training. BMVC 2026

P. Konz, T. Medi, M. Keuper. Unsupervised Anomaly Detection Using Flow Matching on Tabular Data. MIDAS Workshop, ECML-PKDD 2026

T. Medi, A. Rampini, P. Reddy, P. K. Jayaraman, M. Keuper. 3D-WAG: Hierarchical Wavelet-Guided Autoregressive Generation for High-Fidelity 3D Shapes. BMVC 2025

T. Medi, S. Jung, M. Keuper. FAIR-TAT: Improving Model Fairness Using Targeted Adversarial Training. WACV 2025

T. Medi, S. Jung, M. Keuper. TRIX: Trading Adversarial Fairness via Mixed Adversarial Training. arXiv:2507.07768, 2025

T. Medi, J. Tayyub, M. Sarmad, F. Lindseth, M. Keuper. FullFormer: Generating Shapes Inside Shapes. DAGM GCPR 2023

T. Medi, J. Grabinski, M. Keuper. Towards Class-wise Robustness Analysis. OOD Generalization in CV Workshop, ICCV 2023 (Oral)

Technical Skills

Programming	Python
Deep Learning	PyTorch, PyTorch Lightning, TensorFlow; transformers, autoregressive models, flow matching, adversarial training
Scale & Infra	Distributed training with Ray, CUDA, multi-GPU HPC clusters
Engineering	Git/GitHub, reproducible research codebases, academic writing
Languages (spoken)	English (fluent)

Talks

* FullFormer: Generating Shapes Inside Shapes - Invited talk, SFI Visual Intelligence, Norway, Oct 2023.

* Towards Class-wise Robustness Analysis - Oral presentation, ICCV 2023 Workshop, Paris, Oct 2023.

Teaching & Mentoring

* Thesis supervision: supervised ~10 Bachelor's and Master's theses in computer vision, generative AI and ML.

* Teaching assistant: Generative Computer Vision Models (Fall) and Reinforcement Learning (Spring), University of Mannheim.

* Seminars: organised computer vision seminars for Bachelor's and Master's students (2023 - present).

Professional Service

Reviewer: CVPR, ICCV, ECCV, ICLR, WACV, BMVC, TMLR, IEEE TPAMI.

References

Prof. Dr.-Ing. Margret Keuper
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